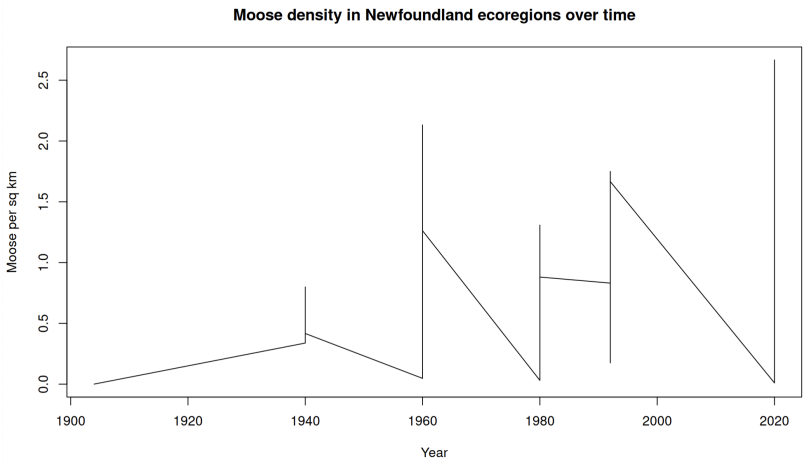
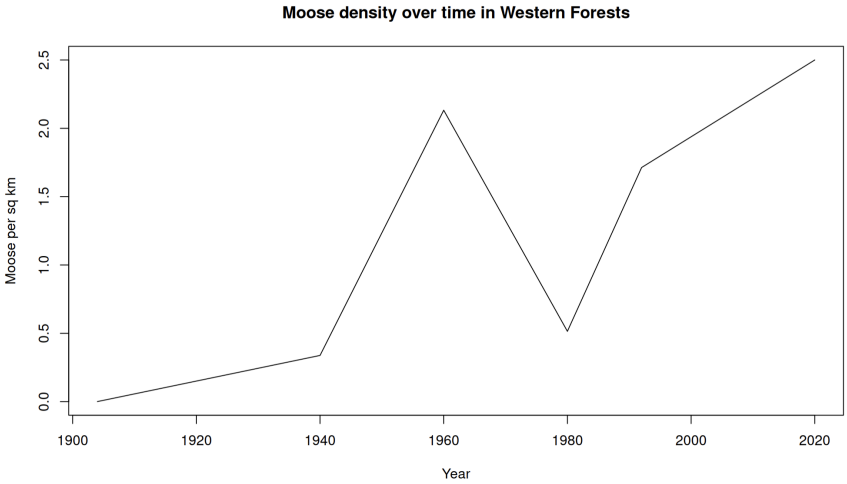
R Assignment
Biology 1002
Alaina Goldsworthy
Memorial University of Newfoundland
February 13th, 2026

Question 5a: 41250L
b: 1904L

Question 7:



Question 8b:



Question 12:

a) The moose browsing pressure varies across ecoregions. The highest average browsing occurred in theWestern Forest and Northern Peninsula Forest, and the lowest browsing occurred within the strait of Belle Isle Barrens.
b) Highest average browsing → Northern Peninsula Forest
Lowest average browsing → Strait of Belle Isle Barrens

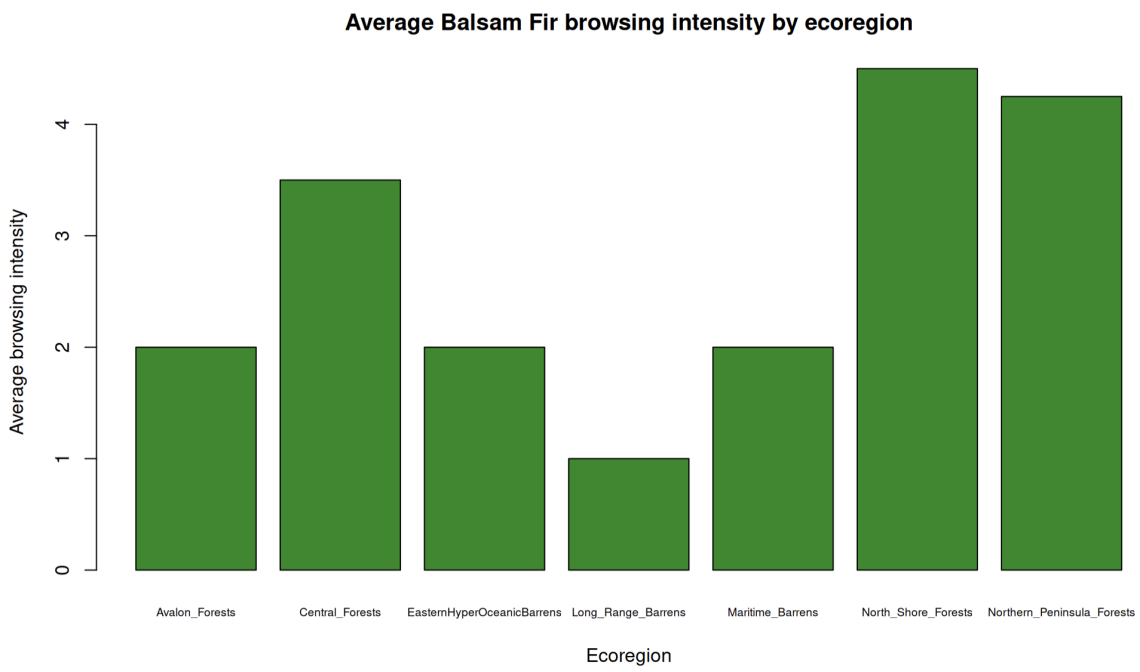
Question 13:

a) The average tree height varies across ecoregions with the shortest average sapling heights occurring in the Western and Northern Peninsula forests and the taller average sapling in areas such as the North Shore and Central Forests.
b) Ecoregions with average tree heights less than 20cm were identified. Two severely browsed ecoregions include the Western Forest and the Northern Peninsula Forest.

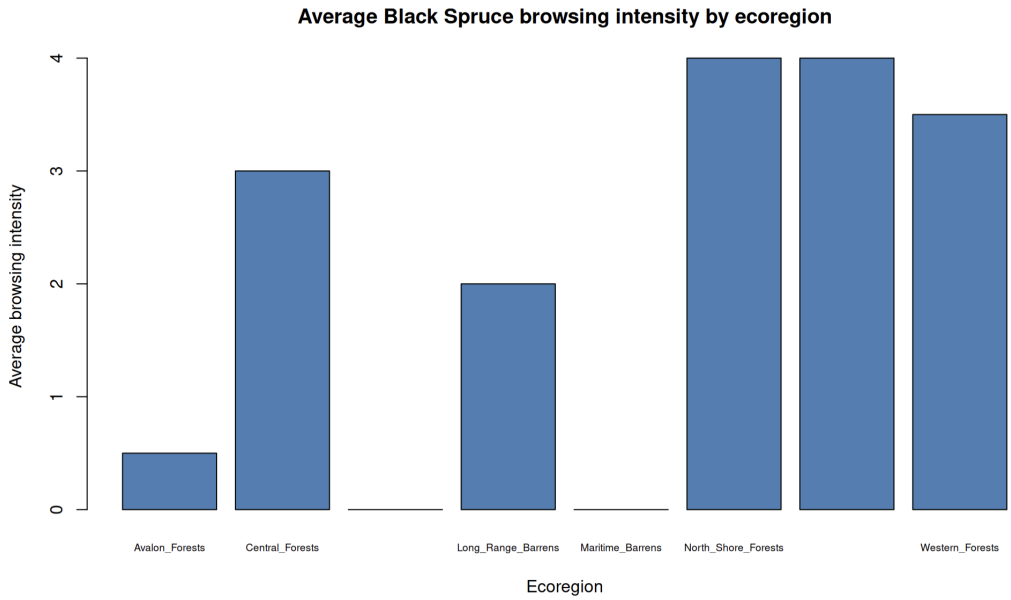
Question 14:

a) The average browsing score varies across tree sapling species. The moose browsed Black Ash (5.00) and Willow (4.31) the most on average and Black Spruce (2.33) the least leaving it with the lowest average browsing pressure.
b) Highest Average Browsing → Black Ash (5.00)
Lowest Average Browsing → Black Spruce (2.33)

Question 16:



Question 17:
a/b)



c) From looking at the graph, Black Spruce shows a lower average browsing intensity than Balsam Fir across most ecoregions. This data indicates that the moose preferentially browse Balsam Fir more than Black Spruce.

Question 18:
No, the same number of tree saplings were not counted in each ecoregion. Areas such as the North Shore Forests (8) had many more saplings sampled than other areas such as Strait of Belle Isle Barrens (1).

Question 19:
No, the same number of tree sapling was not the same for each species. Species such as Balsam Fir (11) were sampled much more frequently than others such as Black Ash (1) which was sampled very few times.

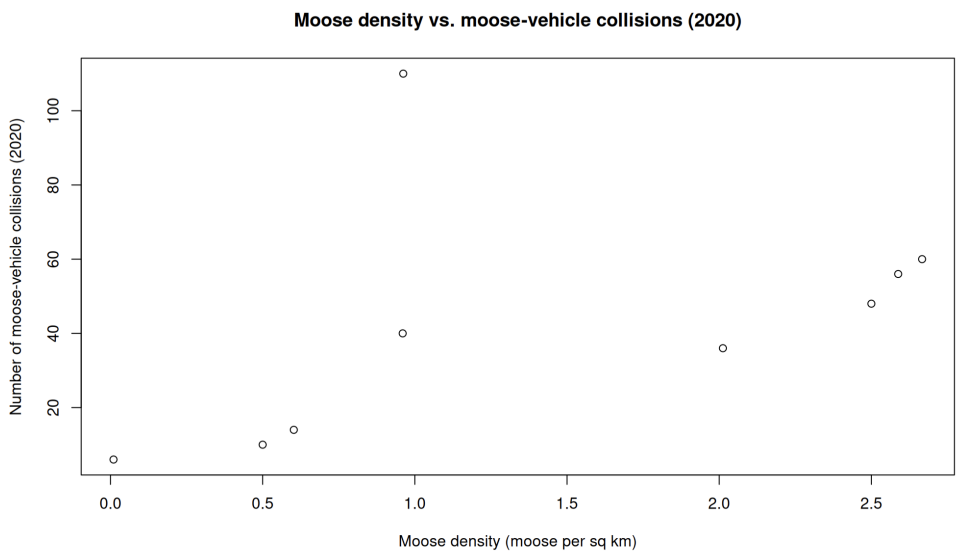
Question 20:

a) The Sapling study dataset is not evenly distributed. There are some ecoregions and species overrepresented while others are underrepresented.
b) Recognizing this bias is important as uneven sampling can lead to a misleading conclusion. By using a biased dataset, it may over/under estimate browsing patterns for certain regions or species.

Question 23:

a) Yes, there is evidence in the graph that supports the hypothesis. When the moose density is lower, browsing varies more among species making it more selective and when the moose density is higher, browsing scores are higher and show more similarities across species becoming less selective.
b) The moose appear to favour Black Ash and Willow the most as they have consistently higher browsing scores and Black spruce the least as it shows consistently lower browsing scores across densities.
c) Black Ash, as it was rarely sampled and only occurred in one region. As a result of this, it does not appear across the full range of moose densities in the figure.

Question 26:

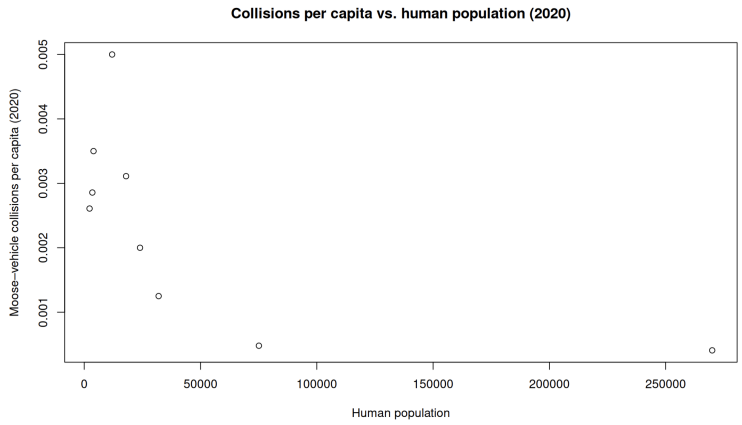


a/b) The most density shows a general positive relationship between the moose density and the number of moose vehicle collisions. Where the ecoregions are higher, moose densities tend to have more collisions. Avalon Forests stands out with very high collisions relative to moose density, this is most likely due to the higher human population and traffic volume.

Question 27:

The highest number of moose vehicle collisions per person occurred in the Northern Peninsula Forest and the Long Range Barrens.

Question 28:



Question 29:

The collisions per capita are higher in ecoregions that have smaller human populations and lower in more densely populated regions. This trend can be supported as rural areas in Newfoundland have more of an overlap between mooses and roadways, this increases the per person risk of moose vehicle collisions compared to more urban areas.